

WOOKYOUNG KIM, Ph.D.

Senior Researcher | Thermal-Fluid Engineering

wookyoung@kimm.re.kr | github.com/WookyoungWoody | linkedin.com/in/wookyoungwoody | Daejeon, Korea



PROFESSIONAL SUMMARY

Thermal-fluid engineer with 5+ years of R&D experience at Korea's national research institute (KIMM). Specialized in thermal management solutions for AI data centers, hydrogen energy systems, and heat pump technology. Extensive hands-on experience in thermal system design, experimental facility construction, performance testing, and data analysis. Spearheading development of multiple engineering software tools (Python/FastAPI/React Native). Track record: 17+ journal papers, 17 patents, 8 registered software programs, 5 technology transfers to industry.

CORE COMPETENCIES

Thermal Management

Data center cooling
Immersion / DLC / Jet cooling
Electronics thermal design

Energy Systems

Heat pump (vapor./ads.)
Hydrogen vaporizer (PCHE)
Low-GWP refrigerant R&D

System Design

Heat exchanger design
Thermal loop construction
High-pressure cryogenic sys.

Testing & Analysis

Performance testing
2-phase flow visualization
VLE measurement & EOS

EXPERIENCE

Senior Researcher | KIMM (Korea Institute of Machinery and Materials) | Mar 2021 - Present

Korea's national research institute for machinery and materials engineering

- PI: Jet-enhanced immersion cooling for next-gen AI data center servers
- PI: Cross-platform thermophysical property app (iOS/Android, React Native + FastAPI)
- Lead: PCHE design for liquid hydrogen vaporizers (-220°C, 100 MPa class)
- Lead: Immersion cooling waste heat utilization and thermal management systems
- Delivered 5 technology transfers to industry partners
- Developed 8 registered engineering software programs (Korea Copyright Commission)

EDUCATION

Ph.D.	Mechanical Engineering	KAIST	2017 - 2021
Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes			
Advisor: Prof. Sung Jin Kim			
M.S.	Mechanical Engineering	KAIST	2015 - 2017
B.S.	Mechanical Engineering	KAIST	2011 - 2015

KEY ACHIEVEMENTS

17+	17	8	5
Journal Papers (10 SCI + 7 KCI)	Domestic Patents	Registered Software Programs	Technology Transfers to Industry

SELECTED PUBLICATIONS

- W. Kim** et al., "Freezing Phenomenon in PCHE for Cryogenic LH2 Vaporizer," Int. J. Heat Mass Transfer (2025)
- W. Kim** and S.J. Kim, "Fundamental issues about pulsating heat pipes," J. Heat Transfer - ASME 143 (2021)
- W. Kim** and S.J. Kim, "Flow behavior effect on pulsating heat pipes," Int. J. Heat Mass Transfer 149 (2020)
- W. Kim** and S.J. Kim, "Reentrant cavities on pulsating heat pipe," Appl. Therm. Eng. 133 (2018)
- J.S. Kim, W. Kim** et al., "Pool boiling of ammonia outside enhanced tubes," Appl. Therm. Eng. 247 (2024)

TECHNICAL SKILLS

Software: Python, JavaScript/TypeScript, C/C++, FastAPI, React, React Native, Git, Docker, Linux

CAD / CAE: INVENTOR, SOLIDWORKS, COMSOL Multiphysics, ANSYS FLUENT, CoolProp, REFPROP

AI / ML: PyTorch, JAX, PINN

Experiment: Thermal loop design & construction (1-phase / 2-phase), Low GWP Refrigerant system, 2-phase flow & heat transfer measurement, High-pressure testing (2000 bar), Flow visualization